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Box target

Abstract

A firearm target is formed from a cardboard substrate. A first target is printed on the cardboard substrate. The cardboard substrate is folded into a box shape. The cardboard substrate is cut to form a first flap and a second flap on a first side of the cardboard substrate. The first flap and second flap include notches. The first flap and second flap are folded over with the notches interlocked. The flaps can be oriented to the bottom of the box shape and weighted for stability. A second target can be mounted to the box shape over the first target.

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Background/Summary

CLAIM TO DOMESTIC PRIORITY (1) The present application is a continuation of U.S. patent application Ser. No. 17/138, 489, now U.S. Pat. No. 11,841,213, filed Dec. 30, 2020, which is a division of U.S. patent application Ser. No. 16/213, 426, now U.S. Pat. No. 10,907,941, filed Dec. 7, 2018, which claims the benefit of U.S. Provisional Application No. 62/597, 351, filed Dec. 11, 2017, which applications are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates in general to firearms, and, more specifically, to a firearm target in the form of a cardboard box.

BACKGROUND OF THE INVENTION

(2) Firearms have a wide variety of uses, including recreation, hunting, and self-defense. Whatever use a person puts a firearm to, an important aspect of firearm ownership is proper firearm training. Target shooting is important to practice and improve shooting skill. Target shooting is also commonly used to prove a minimum competency level or compete against others in organized competitions.

(3) Commonly, shooters wishing to practice will take their firearms to a wilderness or rural area either designated for shooting or remote enough to be safe for shooting. A shooter will set up a target to practice with an area behind the target that is free of other people that could be injured or property that could be damaged by the shooting.

(4) The locales selected for practice shooting may not be set up well for target practice. The shooter may have no option other than to shoot at tin cans or other objects set on the ground, in which case the shooting angle may not be realistic to the situation the shooter is practicing for. The shooter can try tacking a target up on a tree, but then the tree is undesirably damaged by the firing.

(5) Shooting is a favorite American pastime, but setting up a proper target for self-practice remains a challenge. Therefore, a need exists for an improved firearm target.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1a and 1b illustrate a box target;

(2) FIGS. 2a-2e illustrate manufacturing the box target and folding up the box target for distribution;

(3) FIGS. 3a-3c illustrate deploying the box target;

(4) FIGS. 4a-4c illustrate using weight to stabilize the box target;

(5) FIGS. 5a-5e illustrate using flaps of the box target as feet to stabilize the box target; and

(6) FIGS. 6a-6e illustrate using the box target as a substrate for a supplemental target.

DETAILED DESCRIPTION OF THE DRAWINGS

(7) The present invention is described in one or more embodiments in the following description with reference to the figures, in which like numerals represent the same or similar elements. While

the invention is described in terms of the best mode for achieving the invention's objectives, it will be appreciated by those skilled in the art that it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims and their equivalents as supported by the following disclosure and drawings.

(8) FIGS. **1a** and **1b** illustrate a box target **10** for use in practicing firearm shooting. Box target **10** is formed from similar materials and using similar manufacturing processes to common cardboard boxes. Box target **10** is a rectangular box and includes 6 faces like other rectangular cardboard boxes. Side faces **12**, **14**, **16**, and **18** are oriented perpendicular to the ground and offer four different faces that shooter **20** can use as a target. End faces **22** are formed by interlocking flaps at the ends of box target **10** that are used to enclose the box and keep faces **12-18** square. In one embodiment, each side face **12-18** is a rectangle 54.3125 inches tall and 13.125 inches wide and each end faces **22** are a square 13.125 inches to a side. In other embodiments, box target **10** is formed with any suitable dimensions.

(9) Side faces **12-18** can each have different types of targets printed on the outside of box target **10** to give shooter **20** a variety of target options. The targets are optionally printed vertically mirrored so that the top half of box target **10** includes upright targets and the lower half includes upside down targets. The top half of box target **10** is at a good vertical level for target shooting. A shooter can fire at targets on the upper half of box target **10** and then flip the box target over to begin shooting at the unused targets that were previously on the bottom half but are now on the top half. Shooter **20** can aim at the targets on the lower half of box target **10** if desired. Box target **10** is normally printed symmetrically so flipping the box target over provides the same target options. However, some embodiments have different targets printed on top and bottom of box target **10**.

(10) FIG. **1a** shows shooter **20** firing at side face **12** of box target **10**. Side face **12** includes standard $\frac{2}{3}$ scale International Practical Shooting Confederation (IPSC) metric style targets **32**. IPSC targets **32** are a standardized design commonly used in competitions and examinations. Practicing on IPSC targets **32** is desirable so that practice is performed on the same target as used in competition, or so that box target **10** can be used in competition. IPSC targets **32** include various regions marked with the letters A, B, and C. Firing and hitting the "A" regions is the most desirable, while hitting "B" or "C" regions results in a reduced score. IPSC target **32a** is upright on the top half of box target **10**, while IPSC target **32b** is upside down. Shooter **20** can shoot at IPSC target **32a**, and then flip over box target **10** to use IPSC target **32b**, which remains fresh. IPSC targets **32** are provided as two-thirds scale targets due to the dimensional limitation of box target **10**. However, FIGS. **6a-6e** below illustrate one option for attaching a full scale IPSC target to box target **10**. In addition, box target **10** can be made large enough to accommodate a full-scale IPSC target by enlarging faces **12-18** and **22**.

(11) Side face **14** is also seen in FIG. **1a**. Side face **14** includes four diamond targets **42**. Diamond targets **42** each include a plurality of concentric diamond shapes. Shooter **20** can determine the accuracy of a shot by observing which of the concentric diamonds the shot hits closest to. Other target styles can be used in other embodiments, e.g., concentric circles, non-overlapping circles, or any other arbitrary shapes. Any target design can be used on any of the faces of box target **10**. Shooter **20** can turn box target **10**, or move in front of face **14**, to use diamond targets **42** instead of IPSC targets **32**.

(12) FIG. **1B** illustrates box target **10** with faces **16** and **18** oriented toward the viewer. Side faces **16** and **18** include silhouette target **52** split across the two faces. Silhouette target **52** includes discrete regions with numbers indicating a point value received for hitting the particular region. Printing a target design across two adjacent side faces provides a larger overall target for shooter **20**. Edge **54**, which is the corner of box target **10** located between faces **16** and **18**, is oriented toward shooter **20** for firing. Looking at both faces **16** and **18** together results in the appearance of a single target larger than any single side of box target **10**. Any target design can be printed across two adjacent faces to provide a larger target than can be provided on only a single face. Any two

faces can be selected. However, extending a target across the manufacturer's glue joint can create a discontinuity between the two halves of the printed target if the faces of box target **10** are not aligned properly.

(13) FIG. **2a** illustrates cardboard **60** printed and cut for making box target **10**. Printing is not illustrated in FIG. **2a** to better illustrate the physical features of cardboard **60**. However, the side of cardboard **60** that is visible in FIG. **2a** will have targets printed before continuing with gluing and folding as shown in FIGS. **2b-2e**. Printing can be performed before or after cutting and scoring cardboard **60**. The reverse side of cardboard **60**, which is not visible in FIG. **2a**, is usually left blank. The material for cardboard **60** is biodegradable and environmentally friendly single wall board with edge crush test (ECT) grade 32, #3 White Exterior, B Flute with 33/23/33 basis weights. In another embodiment, 42/23/42 basis weights are used. Using heavier weights increases the strength of box target **10** while increasing cost.

(14) Box target **10** can be constructed of corrugated cardboard with any suitable fluting and basis weight. Corrugation lines can be oriented vertically when box target **10** is in use to increase strength. Non-fluted cardboard stock can also be used. Other suitable materials, such as plastic, fabric, metal, or paper are used in other embodiments. The targets are printed in GCMI **74** red for increased visibility, but any suitable color is used in other embodiments.

(15) Cardboard **60** can be cut out using a die to cut every feature required for box target **10** in a single pass. The cut-out features of cardboard **60** include flaps **70**, notches **72** at the ends of the flaps, and tabs **74** in the flaps. In one embodiment, each flap **70** extends eight inches from a respective side face **12-18**. Notches **72** are configured to interlock with notches **72** of an opposing flap **70**, as illustrated below in FIGS. **3b** and **3c**, to keep the flaps closed. In addition, one flap **70** at each end of box target **10** includes an opening **76** cut out of the flap. Each flap **70** also has a perforation **78** cut into a portion of the boundary between the flap and a respective face **12-18**. In one embodiment, perforations **78** are a 3/16 inch× 3/16 inch perforation, meaning each individual cut of the perforation is 3/16 inches long and 3/16 inches apart from adjacent cuts.

(16) Side faces **12-18** each have openings **80** formed at both ends, near flaps **70**. Finger holes **82** are cut out near boundaries between faces, flaps **84** are cut into face **12** near the center of box target **10**, and flap **86** is formed along the edge of face **12** for the length of box target **10**.

(17) In addition to cutting, cardboard **60** undergoes scoring to facilitate proper folding of box target **10**. Scoring of cardboard **60** helps the cardboard bend easier and more reliably in the desired location. Scoring is imparted on cardboard by the manufacturer by pressing down on score lines with an edge that physically compresses the cardboard. The score mark makes the cardboard easier to fold by imparting a pre-bend at the desired locations. In addition, the score marks weaken the cardboard material in a linear fashion so that folds occur cleanly along the score lines when force is applied to fold the cardboard. In one embodiment, the scores for cardboard **60** are 8 point scores.

(18) Flaps **70** have scores **90** along the boundary between the flaps and side faces **12-18**, including through perforations **78**. Scores **92** and **94** are formed through flaps **70** perpendicularly to score **90**. Score **92** is formed at the center of flaps **70** and extends to notch **72**. Score **94** is formed between the two tabs **74** of each flap **70**. Scores **96** are formed at the boundary between each face **12-18** and flap **86**. Scores **98** are formed at the bottom of flaps **84**, and scores **99** are formed at the bottom of tabs **74**. Scores **100** and **102** are formed perpendicularly to scores **96** for folding box target **10** into a manageable size for storage and shipment. The purpose of each cut and score of cardboard **60** is explained in more detail below with reference to how those features aid in assembling, deploying, and using box target **10**.

(19) To begin assembling box target **10** from cardboard **60**, the cardboard is first folded along scores **96** so that flap **86** is disposed behind face **14**. An adhesive is sprayed, rolled, or otherwise disposed on flap **86** or the back of face **14** to attach the flap. The interface between flap **86** and face **14** is referred to as the manufacturer's glue joint. In other embodiments, flap **86** is stapled, sewn, or otherwise attached to face **14**.

(20) FIG. 2b shows cardboard 60 formed into a tube with face 14 over flap 86 to show the relative positions of faces 12-18 and flap 86 when box target 10 is assembled. However, in other embodiments cardboard 60 remains in a flat configuration for gluing of flap 86, as illustrated in FIG. 2c. Cardboard 60 is placed on a flat surface with the printed side oriented down. Face 12 is folded across score 96c so that flap 86 is near score 96b. Then, face 14 is folded across score 96a so that the back of face 14 is on top of flap 86. Scores 96b and 96d are not folded yet, and are now vertically aligned near the center of cardboard 60.

(21) With flap 86 glued to face 14, box target 10 is now ready for use. Box target 10 can be stood up vertically and used as a target. However, normally box target 10 will be packaged for sale rather than immediately used by an end user. FIG. 2d shows beginning to package box target 10 by folding cardboard 60 along scores 102.

(22) Box target 10 is laid flat, with faces 16-18 oriented upward and faces 12-14 oriented down. Faces 12-18 can be in any orientation. Face 14 is the most convenient to print branding, instructions, and other non-functional text on. Having face 14 oriented downward when folding over scores 102 allows the printing on face 14 to be visible when packaging is complete. However, having faces 12 and 14 oriented upward may also be convenient due to score 96d already being oriented upward after gluing flap 86. Moreover, instructions and branding can be printed on any face 12-18 or added as a sticker or insert later. Box target 10 is folded along scores 102 in FIG. 2d. Scores 102 are formed at approximately halfway between the ends of flaps 70 and the centrally located scores 100. In other words, scores 102 are formed at approximately $\frac{1}{4}$ and $\frac{3}{4}$ of the full length of box target 10 including flaps 70.

(23) When completely folded along scores 102, the ends of flaps 70 each extend to near the middle of box target 10 at scores 100. Scores 102 are optionally formed in pairs. A surface 112 between the pairs of scores 102 provides some distance between the two portions of box target 10 on either side of scores 102. The offset provided by surface 112 helps the two sides of the fold to lie flat and parallel to each other. The area of cardboard 60 between scores 102 can be crushed during manufacturing to further aid in folding. In one embodiment, the distance between adjacent scores 102 is 9/16 inches.

(24) Score 100 is also formed as a pair of scores in some embodiments, and cardboard 60 can be crushed between the two scores 100. A surface 114 between scores 100 provides some distance between the two sides of the fold along scores 100, similarly to surface 112 for scores 102. Scores 100 are further apart than scores 102, one inch in some embodiments. Box target 10 is next folded along scores 100 with flaps 70 between the two sides of the fold. Surface 114 is wide enough to allow the full thickness of four stacked flaps 70 between the two sides of the fold.

(25) FIG. 2e shows the fold along scores 100 completed. Box target 10 remains substantially flat, and is folded into a rectangle that is convenient for storage, transportation, and sale. Tape 120 is used to keep box target 10 folded up until sold and ready for use. In other embodiments, box target 10 can be shrink wrapped, clipped, or kept folded using any other suitable mechanism. Box target 10 in FIG. 2e is ready for sale. Any needed branding, marketing information, bar codes, etc., are printed on the visible portion of either face 12 or face 14 to be visible on store shelves without unfolding box target 10. Instructions and more detailed information that is not necessarily viewed prior to purchase can be printed on other surfaces that are not visible in FIG. 2e. A folded box target 10 has a shipping size of approximately 18 inches×26 inches×1 inch and weighs approximately 3 pounds. While one specific folding pattern is taught, any suitable folding pattern is used in other embodiments.

(26) FIGS. 3a-3c illustrate shooter 20 deploying box target 10 after purchase. Shooter 20 initially cuts or removes tape 120 and unfolds box target 10 in the reverse order of steps shown above in FIGS. 2c-2e. After unfolding box target 10 along scores 100 and 102, shooter 20 positions the box target vertically with flaps 70 at the top and bottom as shown in FIG. 3a. Shooter 20 inserts his or her fingers into finger holes 82 to pull box target 10 from flat to tubular.

(27) Box target **10** includes two pairs of finger holes **82**. Both ends of box target **10** include a pair of finger holes so that opening box target **10** is the same action no matter which end is oriented up in FIG. **3a**. Finger holes **82a** and **82b** are formed offset from each other so that the two finger holes do not combine to form a hole extending completely through box target **10** when folded flat. Shooter **20** inserts a finger of his left hand into finger hole **82a** and presses against cardboard on the opposite side of box target **10** rather than having his finger also go through finger hole **82b**. Likewise, the right finger of shooter **20** is inserted through finger hole **82b** and presses against cardboard on the opposite side. Shooter **20** could also open box target **10** with finger hole **82b** on the left and finger hole **82a** on the right. Offset finger holes **82** help shooter **20** easily locate the middle of box target **10**. Finger holes **82a** and **82b** can be formed directly across from each other, but inserting a finger between the layers of cardboard is more challenging without having the opposite side surface to press on through the finger holes.

(28) Finger holes **82** have a $\frac{1}{2}$ -inch radius. Finger hole **82a** is formed directly on score **96b** near the top of side faces **16** and **18**, approximately 1 and $\frac{7}{8}$ inches from flap **70**. Finger hole **82b** is formed lower and on side face **12**, slightly off of score **96d** to ease logistical issues with forming an opening that overlaps the manufacturer's glue joint. Opening **82b** is $\frac{3}{8}$ inches from score **96d** and $\frac{7}{8}$ inches from the nearest score **100**. Finger holes **82** can be any suitable size and at any suitable location on cardboard **60** in other embodiments.

(29) With fingers inserted into finger holes **82**, shooter **20** pulls box target **10** open until side faces **12-18** approximately form a square when viewed from above. Shooter **20** then engages flaps **70** to close box target **10** and maintain the shape of the box target. Two opposing flaps **70** are folded down in FIG. **3b** and interlocked using notches **72**. Notches **72** are formed at the middle of flaps **70** so that opposing notches are aligned with each other when the flaps are folded in. Each flap **70** extends over half the way across box target **10**, so that opposing flaps overlap. Each flap **70** is disposed over the opposite flap on one side of notches **72**, and under the opposite flap on the other side of notches **72**. In some embodiments, the bottoms of all notches **72** are formed to sit approximately at the center of the square formed by side faces **12-18** so that interlocked notches fit together tightly.

(30) In FIG. **3c**, the remaining two flaps **70** are folded over and interlocked using notches **72**. In some embodiments, flaps **70** attached to faces **12** and **16** are folded first to facilitate use of opening **76** with an auxiliary target as shown in FIG. **6c**. However, the flaps can be folded in any order. Interlocking flaps **70** using notches **72** helps box target **10** maintain a box shape. The sides of interlocked notches **72** press against each other to resist forces that would otherwise fold box target **10** along scores **96**. In addition, each flap **70** is held down by an opposite flap **70** because each flap is on top of the other on one side of notch **72**.

(31) With all four flaps **70** on top of box target **10** engaged, shooter **20** flips over the box target and engages flaps **70** on the other end of the box target in the same manner. Box target **10** a cardboard box, now usable as a firearm target. Box target **10** can typically be stood up on one end face **22** after folding flaps **70**, and the box target will stay upright. Bullets fired at box target **10** tear through the cardboard easily without significant energy being transferred from the bullets to the target. Box target **10** easily withstands bullet fire without falling over because the relatively weak cardboard material does not provide significant resistance to the motion of bullets.

(32) On the other hand, wind or unlevel ground could be a problem for keeping box target **10** upright. Flaps **70** on the bottom of box target **10** can be unfolded as shown in FIG. **4a** to provide a mechanism for holding the box target upright. Any kind of weight can be added on top of the bottom flaps **70** to hold down box target **10**. Shooter **20** can use anything found nearby, e.g., logs, sand, or rocks, to weigh down the box target. Shooter **20** can also keep flaps **70** at the bottom of box target **10** closed and add weight into the inside of the box target before closing the top flaps **70**. Another option is for shooter **20** to attach flaps **70** to the ground using stakes.

(33) While any suitable objects with sufficient weight and density can be used to hold flaps **70** onto

the ground, FIG. 4b illustrates a square frame weight **130** that can be sold or provided as an accessory to box target **10**. Weight **130** is a metal bar bent into a square with a length and width slightly larger than the square formed by sides **12-18**. In one embodiment, weight **130** is a 5-pound $\frac{5}{8}$ " diameter steel rod bent into a 15-inch×15-inch square. Shooter **20** can dispose weight **130** over the top of box target **10**, and slide the weight down the box target until the weight rests on the bottom flaps **70** as shown in FIG. 4c. Weight **130** presses down on all four flaps **70** on the bottom end of box target **10** by the force of gravity. The weight on flaps **70** keeps box target **10** in place due to the attachment of flaps **70** to faces **12-18**. In addition, the square shape of weight **130** helps keep faces **12-18** perpendicular to each other without flaps **70** interlocked via notches **72**. When shooter **20** desires to flip box target **10** to have fresh targets to shoot at, the weighed down flaps in FIG. 3c are folded with notches **72** engaged to become the top flaps. Top flaps in FIG. 3c are opened to have weight added at the bottom of the box target.

(34) As an alternative to weighing down flaps **70**, the lower flaps can be folded into legs to prop up box target **10**. FIGS. 5a-5e illustrate creating support legs out of flaps **70**. The process is illustrated as being performed on the top flaps **70** because assembling the flaps into legs near arm level is more convenient than at ground level. The top flaps **70** can be folded into legs as illustrated, and then box target **10** flipped to have the folded flaps at the bottom of box target **10**.

(35) In FIG. 5a, perforation **78** of a flap **70** is torn to partially separate the flap from a corresponding side face **12-18**. Flap **70** is folded along scores **92** and **94** to give flap **70** an extension upward in front of the side face **12-18**. In one embodiment, flap **70** is folded at a ninety-degree angle at score **92** and then folded back onto itself at score **94**, as illustrated in FIG. 5b. Score **94** can be folded either toward perforation **78** or away from the perforation.

(36) In addition to folding flap **70** along scores **92** and **94**, tabs **74** are folded down toward the corresponding face **12-18** along scores **99**. The two tabs **74** can be folded in opposite directions from each other as shown in FIG. 5b, or in the same direction as shown in FIG. 5c, irrespective of the folding direction along score **94**. The folded-back tabs **74** are inserted into opening **80** as shown in FIG. 5d. The two tabs **74** folded into a single opening **80** apply counter pressure to each other to help hold the tabs into the opening. Tabs **74** also have heads **140** with corners that resist forces pulling the tabs out of opening **80**. Heads **140** help keep tabs **74** from being pulled out of opening **80** during use of box target **10**.

(37) In one embodiment, tab **74a** is cut into flap **70** with a greater length than tab **74b**. With tabs **74a** and **74b** folded the same direction as shown in FIG. 5c, tab **74a** extends through the hole left by tab **74b** and then downward. Tab **74a** is slightly longer to make up for the length of the tab that extends through flap **70**, thereby putting the heads of both tabs **74** at approximately the same length through opening **80**. Either tab **74** can be made longer, and shooter **20** is instructed that the longer tab should be extended through the opening of the shorter tab.

(38) The process in FIGS. 5a-5d is repeated for all four flaps **70** on one end of box target **10** to complete the full set of four legs. Then, box target **10** is flipped so that the folded flaps **70** are at the bottom end of the box target as shown in FIG. 5e. Tabs **74** inserted into openings **80** maintain the fold of flap **70** upward. Any force on box target **10**, for instance from blowing wind, results in one or more faces **12-18** in the downwind direction pressing against the portion of flaps **70** extending upward. The force is converted downward into the ground by flaps **70**, which extend outward from faces **12-18** to increase leverage. The ground resists the force to keep box target **10** oriented upright. Using flaps **70** as fold-out feet allows box target **10** to be self-supported. No added frames or stakes are needed. The flap **70** feet help support the target on uneven ground or in breezes. The fold-out feet are usually sufficient in up to around 5 mile per hour (mph) winds. The accessory weight **130** is usually sufficient for up to around 12 mph winds.

(39) FIG. 6a shows a full size IPSC target **150** that is an accessory to box target **10**. The target printed on IPSC target **150** is similar to $\frac{3}{4}$ IPSC target **32** but larger. In one embodiment, IPSC target **150** has a width of 46 centimeters and a height of 76 centimeters. Like cardboard **60**, IPSC

target **150** is formed from a cardboard material in some embodiments and has various features cut and scored. The cut features include a tab **152** on the bottom of target **150** and perforations **154** between head **156** and flaps **160** of the target. In one embodiment, perforations **154** are a $\frac{1}{8}$ inch by $\frac{1}{8}$ inch. One of the flaps **160** includes a tab **162** cut into the flap. Score lines are formed at the bottom of flaps **160** for folding the flaps down, and where tab **162** is attached to the rest of flap **160**.

(40) Target **150** is installed onto box target **10** as illustrated in FIGS. **6b-6e**. Shooter **20** pushes in flap **84** and inserts tab **152** of target **150** into the opening of flap **84**. In one embodiment, tab **152** on target **150** is a half-circle with a radius of 1.5 inches, and flap **84** is 3.5 inches across by 1.0 inches tall. The bottom of target **150** around tab **152** rests on the bottom of the flap **84** opening to support the weight of the accessory target. Tab **152** extends behind side face **12** of box target **10**, which keeps the bottom of full size IPSC target **150** close to box target **10**.

(41) FIG. **6c** shows the top of target **150** pushed back so that target **150** is flat on side face **12** with tab **152** still in the opening of flap **84**. Box target **10** has flaps **70** over side faces **12** and **16** already folded down and interlocked. Opening **76** is visible in the flap **70** toward target **150**. Perforations **154** are broken if necessary, and flaps **160** are folded down onto flap **70**. Tab **162** is folded down and inserted into opening **76**. As with tabs **74**, tab **162** includes a head to help keep the tab in opening **76**. In some embodiments, two openings **76** are formed in flap **70**, and both flaps **160** of target **150** include tabs **162**.

(42) In FIG. **6d**, the two remaining flaps **70** of box target **10** are folded down on top of flaps **160** of target **150**. Flaps **70** are interlocked over flaps **160** to help keep all the flaps in place. Flap **70** over tab **162** helps keep the target **150** tab in opening **76**. In other embodiments, tab **162** is not used. Flaps **160** can be inserted between, and removed from, flaps **70** without having to undo flaps **70** of box target **10**. Tab **162** helps keep flaps **160** in place between flaps **70** when, e.g., wind blows the back of target **150**, but is not necessary for use of target **150**.

(43) Target **150** is now installed and ready for use as shown in FIG. **6e**. Having target **150** as a separate accessory allows a full size IPSC target to be used even though box target **10** has insufficient surface area for a full size IPSC target. Accessory targets also extend the life of box target **10** by allowing more exchanging of used targets. Box target **10** by itself can be flipped to use the bottom target, and turned 90 degrees to use another face, but eventually all targets on box target **10** will have too many bullet holes to be useful. However, box target **10** remains structurally sound long after all targets have been used.

(44) A used target **150** full of bullet holes can be changed out for a fresh accessory target without any bullet holes whenever desired. Any number of fresh targets can be used time after time as long as box target **10** remains standing. Even when the underlying targets of box target **10** collects an incredible number of bullet holes, the structural strength of the box target will remain sufficient to hold an accessory target **150**.

(45) Target **150** can be any suitable target, whether wider than a side face **12-18** of box target **10** or not. Targets printed on target **150** can include novelty target such as animals or zombies. In some embodiments, additional flaps **84** are formed at different heights on box target **10** for use of different heights of accessory targets. An accessory target can extend down to the bottom half of box target **10**, or may only take up the top quarter of the box target.

(46) Box Target **10**, with or without accessory targets **150**, is useful for target practice of pistol and rifle armament, as well as airsoft, pellet guns, and BB guns. Most airsoft plastic beads pass through the front of box target **10** but do not have enough energy to pass through the back. Thus, airsoft pellets are trapped within box target **10** and can be collected for reloading. The thickness of cardboard **60** can be customized based on expected projectile energy so that a specific combination of firearm and pellet being used results in pellets collected within the box rather than bouncing off the front or going all the way through the back. In some embodiments, the bottom of box target **10** includes a corner pour hole so that captured pellets can be funneled into a container for reuse.

(47) While one or more embodiments of the present invention have been illustrated in detail, the skilled artisan will appreciate that modifications and adaptations to those embodiments may be made without departing from the scope of the present invention as set forth in the following claims.

Claims

1. A firearm target, comprising: a cardboard substrate folded and glued into a tube including four flat sides of approximately equal size, wherein each flat side is connected to two adjacent flat sides to form the tube; a plurality of cuts with one of the cuts formed between each adjacent pair of the flat sides at a first end and second end of the tube to provide a flap adjacent to each end of each of the flat sides; a first score line formed in the cardboard substrate between each pair of adjacent flat sides to allow the tube to be folded flat; a second score line formed in the cardboard substrate perpendicularly to the first score lines across an entire width of the cardboard substrate at a quarter of the distance from the first end of the tube to the second end of the tube to allow the first end of the tube to be folded to a center of the tube; a third score line formed in the cardboard substrate perpendicularly to the first score lines across the entire width of the cardboard substrate at a quarter of the distance from the second end of the tube to the first end of the tube to allow the second end of the tube to be folded to the center of the tube; and a pair of fourth score lines formed in the cardboard substrate perpendicularly to the first score lines across the entire width of the cardboard substrate near the center of the tube to allow the tube to be folded in half with the first end of the tube and the second end of the tube internal to the central fold.
2. The firearm target of claim 1, wherein each flap includes a notch centered along an edge of the flap opposite a respective flat side.
3. The firearm target of claim 1, further including a weight disposed around the tube and on each of the flaps of the first end of the tube.
4. The firearm target of claim 1, wherein each of the flaps includes a score line oriented perpendicularly to the cuts and formed across an entire width of each flap, and wherein each of the flaps includes a perforation along the score for half of the entire width of each flap.
5. The firearm target of claim 4, wherein each of the flaps further includes a pair of perforated tabs formed adjacent to the perforation.
6. A firearm target, comprising: a cardboard substrate folded and glued into a tube including a plurality of flat sides, wherein each flat side is connected to two adjacent flat sides to form the tube; and a plurality of cuts with one of the cuts formed between each adjacent pair of the flat sides at a first end and second end of the tube to provide a flap adjacent to each end of each of the flat sides; wherein each of the flaps includes a score line oriented perpendicularly to the cuts and formed across an entire width of each flap; and wherein each of the flaps includes a perforation along the score for half of the entire width of each flap.
7. The firearm target of claim 6, wherein the cardboard substrate includes a first score line between each pair of adjacent flat sides to allow the tube to be folded flat.
8. The firearm target of claim 6, wherein each flap includes a notch centered along an edge of the flap opposite a respective flat side.
9. The firearm target of claim 6, further including a weight disposed around the tube and on each of the flaps of the first end of the tube.
10. The firearm target of claim 6, wherein each of the flaps further includes a pair of perforated tabs formed adjacent to the perforation.
11. A firearm target, comprising: a cardboard substrate folded and glued into a tube including four flat sides of approximately equal size, wherein each flat side is connected to two adjacent flat sides to form the tube; a first score line formed in the cardboard substrate between each pair of adjacent flat sides to allow the tube to be folded flat; a second score line formed in the cardboard substrate perpendicularly to the first score lines across an entire width of the cardboard substrate at a quarter

of the distance from a first end of the tube to a second end of the tube to allow the first end of the tube to be folded to a center of the tube; a third score line formed in the cardboard substrate perpendicularly to the first score lines across the entire width of the cardboard substrate at a quarter of the distance from the second end of the tube to the first end of the tube to allow the second end of the tube to be folded to the center of the tube; and a pair of fourth score lines formed in the cardboard substrate perpendicularly to the first score lines across the entire width of the cardboard substrate near the center of the tube to allow the tube to be folded in half with the first end of the tube and the second end of the tube internal to the central fold.

12. The firearm target of claim 11, wherein the cardboard substrate includes a cut between each adjacent pair of the flat sides at a first end of the tube to provide a flap adjacent to each of the flat sides.

13. The firearm target of claim 12, wherein each flap includes a notch centered along an edge of the flap opposite a respective flat side.

14. The firearm target of claim 11, further including a weight disposed around the tube and on each of the flaps.

15. The firearm target of claim 11, wherein each of the flaps includes a score line oriented perpendicularly to the cuts and formed across an entire width of each flap, and wherein each of the flaps includes a perforation along the score for half of the entire width of each flap.
